

UNIT-IV

part B

Linear Motion

1. Two trains A and B leave the same station on parallel lines. Train A starts with a uniform acceleration of $\frac{1}{6} \text{ m/s}^2$ and attains a speed of 24 km/h . When steam is reduced to keep the speed constant, train B leaves after 40 s with uniform acceleration of $\frac{1}{3} \text{ m/s}^2$ to attain a maximum speed of 48 km/h . When will it overtake A? (May 2015, Dec. 2017)

Solution:

The motion of train A:

Initial velocity, $u = 0$,

acceleration, $a = \frac{1}{6} = 0.166 \text{ m/s}^2$

Final velocity, $v = 24 \text{ km/h}$

$$= \frac{24 \times 10^3}{60 \times 60} = 6.67 \text{ m/sec}$$

Time required to attaining
a speed of 24 km

$$v = u + at$$

$$\Rightarrow 6.67 = 0 + 0.166t$$

$$t = \frac{6.67}{0.166}$$

$$= 40 \text{ sec.}$$

Distance travelled by the train A in 40 sec } $S = ut + \frac{1}{2}at^2$

$$= 0 + \frac{1}{2}(0.166)(40)^2$$

$$= 133.33 \text{ m}$$

Distance travelled by train A in 't' sec = $133.33 + 6.67(t - 40)$

$$S_A = 133.33 + 6.67t - 266.8$$

$$= 6.67t - 133.47 \quad \text{--- (1)}$$

The motion of train B:

$$u = 0,$$

$$a = \frac{1}{3} = 0.33 \text{ m/sec}^2$$

$$V = 48 \text{ kmph}$$

$$= \frac{48 \times 10^3}{60 \times 60} = 13.34 \text{ m/sec}$$

Time taken by train B, to attain a speed of 48 kmph } $V = u + at$

$$13.34 = 0 + 0.33t$$

$$t = \frac{13.34}{0.33} + 40$$

$$= 40 + 40 = \underline{\underline{80 \text{ sec}}}$$

Train B starts 40 sec after the start of train A }

Distance travelled by train B in 40 sec } $S_B = ut + \frac{1}{2}at^2$

$$= 0 + \frac{1}{2}(0.33)(40)^2$$

$$= \underline{\underline{266.67 \text{ m}}}$$

Distance covered by train B, $S_B = 266.67 + 13.34(t - 80)$
 $= 13.33t - 800 \quad (2)$

Compare train A train B distance
 covered } ,

$$6.67t - 133.47 = 13.33t - 800$$

$$-13.33t + 6.67t = -800 + 133.47$$

Train B will overtake A } $t = 100 \text{ sec}$
 at the time of

2. A bullet of mass 81 gm moving with a velocity of 300 m/s is fired into a log of wood and it penetrates to a depth of 10 cm. If the bullet moving with the same velocity were fired into a similar piece of wood 5 cm thick, with what velocity would it emerge? Find also the force of resistance assuming it to be uniform. (Jan. 2016)

Solution:

Case 1:

Initial velocity $u = 300 \text{ m/sec}$,

Thickness of wood $S_1 = 10 \text{ cm} = 0.1 \text{ m}$

Case 2:

Thickness of wood, $S_2 = 5 \text{ cm} = 0.05 \text{ m}$

Final velocity, $v = 0$;

$$v^2 = u^2 + 2as$$

Case 1:

$$0 = (300)^2 + 2a(0.1)$$

$$= 90,000 + 0.2a$$

$$0.2a = -90,000$$

$$a = \frac{-90,000}{0.2} = -4,50,000 \text{ m/sec}^2$$

Case 2:

$$v^2 = u^2 + 2as$$

$$= (300)^2 + 2(-4,50,000)(0.05)$$

$$= 90,000 - 4,50,000 = \cancel{4,50,000}$$

$$v = \frac{\cancel{600}}{\cancel{2+2.13}} \text{ m/sec} = 212.13 \text{ m/sec}$$

$$= \underline{\underline{\cancel{600} \text{ m/sec}}}$$

3. A train starts from rest and attains velocity of 45 kmph in 2 min, with uniform acceleration. calculate, (i) acceleration (ii) distance travelled in this time, 2 min. (iii) time required to reach a velocity 36 kmph. (Dec. 2018)

Solution:

Given:

Initial velocity, $u = 0$

Final velocity, $v = 45 \text{ kmph} = \frac{45 \times 10^3}{60 \times 60} = 12.5 \text{ m/sec}$

Time, $t = 2 \text{ min.} = 2 \times 60 = 120 \text{ sec.}$

(i)

$$v = u + at$$

$$12.5 = 0 + a(120)$$

$$120a = 12.5$$

$$a = \frac{12.5}{120} = 0.104 \text{ m/sec}^2$$

$\therefore$ acceleration, $a = 0.104 \text{ m/sec}^2$

(ii) Distance travelled, $s = ut + \frac{1}{2}at^2$

$$= 0(120) + \frac{1}{2}(0.104)(120)$$

$$= 748.8 \text{ m}$$

(iii) Time required to reach velocity 36 kmph,

$$v = 36 \times \frac{10^3}{60 \times 60} = 10 \text{ m/sec}$$

$$v = u + at$$

$$10 = 0 + (0.104)t$$

$$t = \frac{10}{0.104} = \underline{\underline{96.15 \text{ sec}}}$$

4. A motor car takes 10 seconds to cover 20 m and another 15 seconds to cover another 40 m. Find the initial velocity and the constant acceleration. Also determine the velocity at the end of 15 seconds. (May 2016)

Solution:

Given:

$$s_1 = 20 \text{ m}$$

$$t_1 = 10 \text{ sec.}$$

$$s_2 = 40 \text{ m}$$

$$t_2 = 15 \text{ sec.}$$

$$s_1 = ut_1 + \frac{1}{2}at_1^2$$

$$20 = 10u + \frac{1}{2}a(10)^2$$

$$20 = 10u + 50a$$

$\div$ by 10

$$2 = u + 5a$$

$$u + 5a = 2 \quad \text{--- (1)}$$

$$s_2 = ut_2 + \frac{1}{2}at_2^2$$

$$40 = 15u + \frac{1}{2}a(15)^2$$

$$40 = 15u + \frac{225a}{2}$$

$$40 = 15u + 112.5a$$

$$15u + 112.5a = 40$$

$\div$ by 5

$$3u + 22.5a = 8 \quad \text{--- (2)}$$

$$\textcircled{1} \times 3 \Rightarrow \underline{3u + 15a = 6}$$

$$\textcircled{2} \Rightarrow \underline{3u + 22.5a = 8}$$

$$+ 7.5a = + 2$$

$$a = \frac{2}{7.5} = 0.266 \text{ m/sec}^2$$

Sub. in ①,

$$u + 5(0.266) = 2$$

$$u = 2 - 1.33 \\ = 0.67 \text{ m/sec}$$

$$\begin{aligned} \text{Final velocity, } v &= u + at \\ &= 0.67 + 0.266(15) \quad | t = 15 \text{ sec} \\ &= \underline{4.66 \text{ m/sec}} \end{aligned}$$

D'Alembert's principle

1. A motorist, travelling at a speed of 70 kmph suddenly applies brakes and halts after skidding 50 m. Determine
- (a) The time required to stop the car and
 - (b) The co-efficient of friction between the tyres and the road. (NOV. 2016)

Solution:

$$\begin{aligned} u &= 70 \text{ kmph} \\ &= \frac{70 \times 10^3}{60 \times 60} = 19.44 \text{ m/sec}, \end{aligned}$$

(a) $v = 0,$
 $s = 50 \text{ m}$

$$v^2 = u^2 + 2as$$

$$\begin{aligned} 0 &= (19.44)^2 + 2a(50) \\ &= (19.44)^2 + 100a \end{aligned}$$

$$100a = -(19.44)^2$$

$$a = \frac{(19.44)^2}{100} = -3.8 \text{ m/sec}^2$$

$$v = u + at$$

$$0 = 19.44 + (-3.8)(t)$$

$$3.8t = 19.44$$

$$t = \frac{19.44}{3.8} = 5.11 \text{ Sec}$$

Time required to stop the car, $t = 5.11 \text{ Sec}$

(b)

D'Alembert's principle,

$$\Sigma F = ma$$

$$-F = \frac{W}{g} a$$

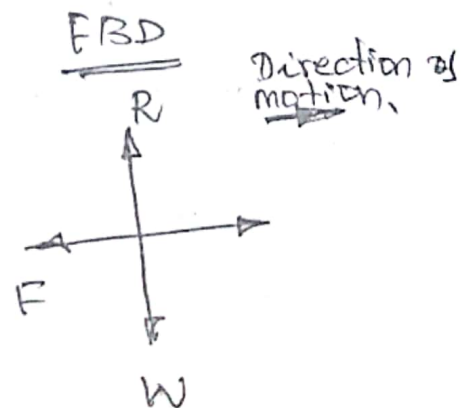
$$- \mu W = \frac{W}{g} (-3.8)$$

$$\cancel{f} \mu \cancel{W} = \cancel{f} \frac{\cancel{W}}{9.81} (3.8)$$

$$\mu = \frac{3.8}{9.81}$$

$$\mu = 0.387$$

$$\therefore \underline{\mu = 0.39}$$



$$\Sigma F_y = 0; R - W = 0$$

$$R = W$$

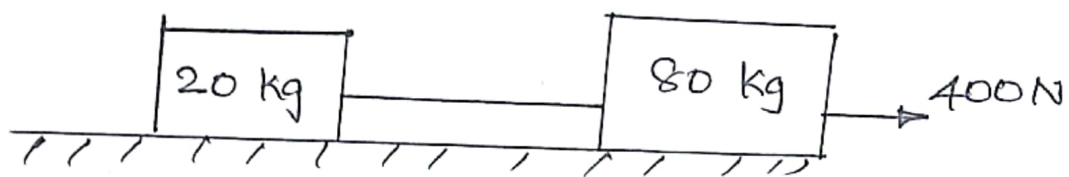
$$F = \mu R$$

$$F = \mu W$$

Coefficient of friction between tyres and the road, $\mu = 0.39$

Note: This problem can be solved by Impulse momentum method

2. Two bodies of masses 80 kg and 20 kg are connected by a thread along a rough horizontal surface under the action of a force 400 N applied to the first body of mass kg as shown in Fig. The coefficient of friction between the sliding surfaces of the bodies and plane is 0.3. Determine the acceleration of the two bodies and the tension in the thread using D'Alembert principle. (May 2019)



Solution:

Given $\mu = 0.3$,
 $m_1 = 20 \text{ kg}$,
 $m_2 = 80 \text{ kg}$,

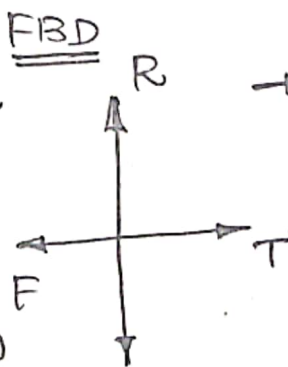
Consider 20 kg,

D'Alembert's principle,

$$\sum F = ma$$

$$T - F = 20a$$

$$T - 58.86 = 20a \quad \text{--- (1)}$$

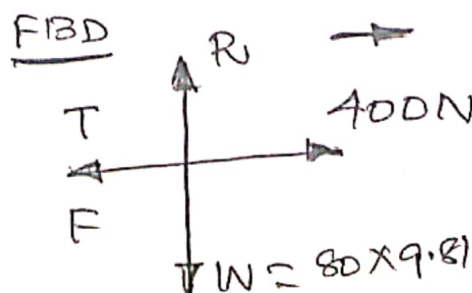


$$W = 20 \times 9.81 = 196.2 \text{ N}$$

Direction of motion.

$$\begin{aligned} \sum F_y &= 0 \\ R - W &= 0 \\ R &= W = 196.2 \text{ N} \\ F &= \mu R \\ &= 0.3 \times 196.2 \\ &= 58.86 \text{ N} \end{aligned}$$

Consider 80 kg,



$$W = 80 \times 9.81 = 784.8 \text{ N}$$

$$\Sigma F = ma$$

$$400 - T - F = ma$$

$$400 - T - 235.44 = 80a$$

$$-T + 164.56 = 80a \quad \text{--- (2)}$$

By solving (1) & (2),

$$\textcircled{1} + \textcircled{2} \Rightarrow \cancel{T} - 58.86 = 20a$$

$$\cancel{-T} + 164.56 = 80a$$

$$105.7 = 100a$$

$$\text{acceleration of bodies, } a = \frac{105.7}{100} = \underline{1.057 \text{ m/sec}^2}$$

Sub. in (1),

$$T - 58.86 = 20(1.057)$$

$$T = 21.14 + 58.86$$

$$= \underline{80 \text{ N}}$$

$\therefore$ Tension in the thread, $T = 80 \text{ N}$

$$\Sigma F_y = 0$$

$$R - W = 0$$

$$R = W = 784.8 \text{ N}$$

$$F = \mu R$$

$$= 0.3 \times 784.8 \text{ N}$$

$$= \underline{235.44 \text{ N}}$$

3. Two blocks of mass $m_A = 280 \text{ kg}$, $m_B = 420 \text{ kg}$ are joined by an inextensible cable as shown in fig. Assuming the pulley is frictionless and $\mu = 0.3$ between the block A and surface. The system is initially at rest. Find the
- acceleration of block A,
 - velocity after it had moved 3.5 m,
 - Velocity after 1.5 sec. (Dec. 2017)

Solution:

Given:

$$m_A = 280 \text{ kg},$$

$$m_B = 420 \text{ kg},$$

$$\mu = 0.3$$

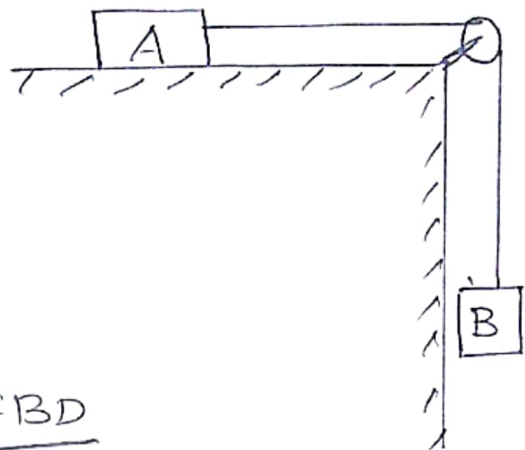
Consider Block A:

D'Alembert's principle,

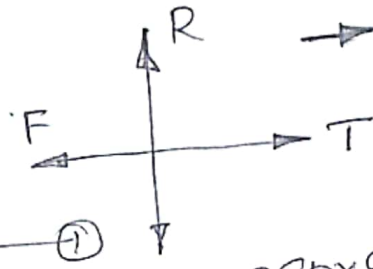
$$\Sigma F = ma$$

$$T - F = 280a$$

$$T - 824.04 = 280a \quad \text{--- (1)}$$



FBD



$$W = 280 \times 9.81$$
$$W_A = 2746.8 \text{ N}$$

$$\Sigma F = 0$$

$$R - W_A = 0$$

$$R = W_A$$
$$= 2746.81 \text{ N}$$

$$F = \mu R$$
$$= 0.3 \times 2746.8$$
$$= 824.04 \text{ N}$$

Consider Block B:

$$\Sigma F = ma$$

$$-T + 4120.2 = 420a \quad \text{--- (2)}$$

① + ②,

$$-824.04 = 280a$$

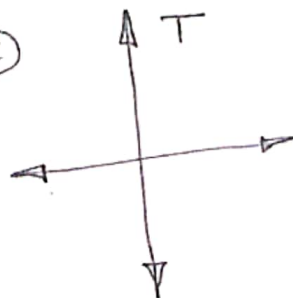
$$-T + 4120.2 = 420a$$

$$3296.16 = 700a$$

$$a = \frac{3296.16}{700} = 4.70 \text{ m/sec}^2$$

(a) acceleration of block, $A = 4.70 \text{ m/sec}^2$

FBD



$$W_B = 420 \times 9.81$$

$$= 4120.2 \text{ N}$$

↓
Direction of motion

Sub. in (1),

$$T - 824.04 = 280a$$

$$T = 280(4.70) + 824.04$$

$$T = \underline{2140.04 \text{ N}}$$

(b) Velocity after it moved, $s = 3.5 \text{ m}$,

$$a = 4.70 \text{ m/sec}^2, u = 0,$$

$$v^2 = u^2 + 2as$$

$$= 0 + 2(4.70)(3.5)$$

$$= 32.9$$

$$\therefore v = \sqrt{32.9} = \underline{5.74 \text{ m/sec}}$$

(c) Velocity after 1.5 sec.,

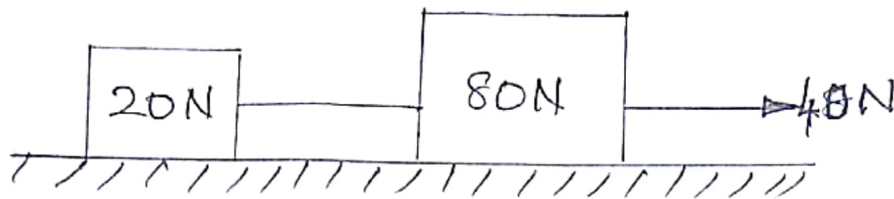
$$v = u + at$$

$$= 0 + (4.70)(1.5)$$

$$= \underline{7.05 \text{ m/sec}}$$

Work Energy Method

1. Two weights 80 N and 20 N are connected by a thread and move along a rough horizontal plane under the action of a force 40 N, applied to the first weight of 80 N as shown in Fig. The co-efficient of friction between the sliding surfaces of the weights and the plane is 0.3. Determine the acceleration of the weights and the tension in the thread using Work-energy equation. (Dec. 2018)



Solution:

Given:

$$\mu = 0.3; u = 0;$$

consider block of 80N;

Work Energy Equation

$$\Sigma F \times S = \frac{W}{2g} (V^2 - u^2)$$

$$(40 - T - F)S = \frac{80}{2 \times 9.81} (V^2 - 0^2)$$

$$(40 - T - F)S = \frac{80}{2 \times 9.81} V^2$$

$$(40 - T - 24)S = \frac{80}{2 \times 9.81} V^2$$

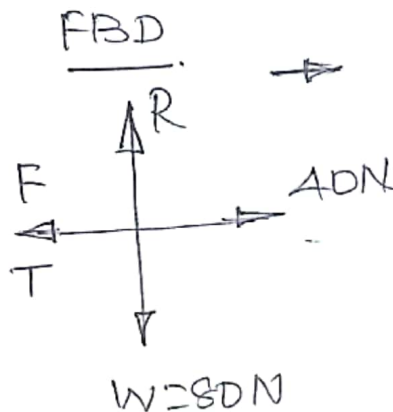
$$(-T + 16)S = 4.077 V^2 \quad \text{--- (1)}$$

Consider block of 20N

$$\Sigma F \times S = \frac{W}{2g} (V^2 - u^2)$$

$$(T - F)S = \frac{20}{2 \times 9.81} (V^2 - 0^2)$$

$$(T - 6)S = 1.01 V^2 \quad \text{--- (2)}$$



$$\Sigma F_y = 0$$

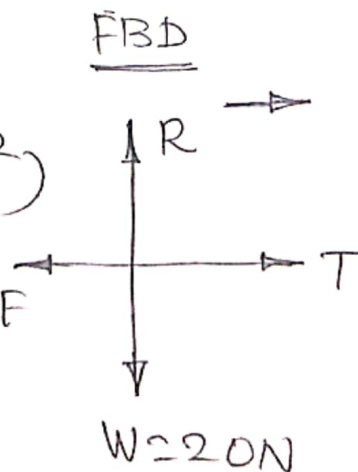
$$R - 80 = 0$$

$$R = 80N$$

$$F = \mu R$$

$$= 0.3 \times 80$$

$$= \underline{\underline{24N}}$$



$$\Sigma F_y = 0$$

$$R - W = 0$$

$$R = W$$

$$= 20N$$

$$F = \mu R$$

$$= 0.3 \times 20$$

$$= \underline{\underline{6N}}$$

$$\frac{\textcircled{1}}{\textcircled{2}} \Rightarrow \frac{(-T+16)\cancel{s}}{(T-6)\cancel{s}} = \frac{4.077\cancel{V^2}}{1.01\cancel{V^2}}$$

$$\frac{-T+16}{T-6} = \frac{4.077}{1.01}$$

$$\frac{-T+16}{T-6} = 4.036$$

$$-T+16 = 4.036(T-6)$$

$$-T - 4.036T = -24.216 - 16$$

$$-T(1+4.036) = -40.216$$

$$T(5.036) = 40.216$$

$$T = \frac{40.216}{5.036} = \underline{\underline{8\text{ N}}}$$

∴ Tension in the thread, $T = 8\text{ N}$

sub. in ①,

$$(-T+16)s = 4.077V^2$$

$$(-8+16)s = 4.077V^2$$

$$8s = 4.077V^2$$

②

$$8\cancel{s} = 4.077(2\cancel{g\cancel{s}})$$

$$8 = 8.154a$$

$$a = \frac{8}{8.154} = 0.98\text{ m/sec}^2$$

acceleration of weights, $a = \underline{\underline{0.98\text{ m/sec}^2}}$

2. In a police investigation of tyre marks, it was concluded that a car while in motion along a straight level road skidded for a total of 60 m after the brakes were applied. If co-efficient of friction between the tyres and the pavement is estimated as 0.5. What was the probable speed of the car just before the breaks were applied. (Sep. 2020)

Solution:

Given:

$$S = 60 \text{ m},$$

$$V = 0,$$

$$\mu = 0.5$$

$$\Sigma F_y = 0$$

$$R - W = 0,$$

$$R = W,$$

$$F = \mu R,$$

$$F = \mu W,$$

Work Energy equation,

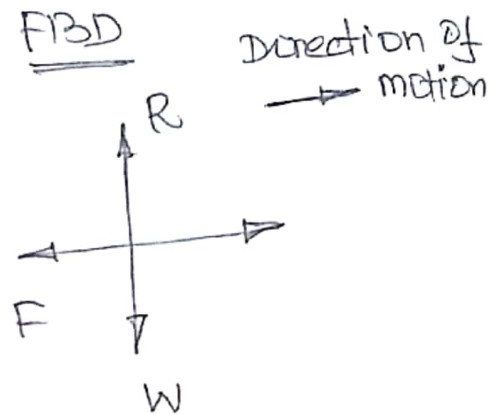
$$\Sigma F \times S = \frac{W}{2g} (V^2 - u^2)$$

$$-F \times 60 = \frac{W}{2 \times 9.81} (0 - u^2)$$

$$-F \times 60 = -\frac{W u^2}{2 \times 9.81}$$

$$+ \mu W \times 60 = \frac{W u^2}{2 \times 9.81}$$

$$0.5 \times 60 = \frac{u^2}{2 \times 9.81}$$



$$F = \mu W$$

$$\begin{aligned}
 u &= 24.261 \text{ m/sec,} \\
 &= \frac{24.261 \times 60 \times 60}{1000} \\
 &= \underline{\underline{87.34 \text{ Kmph}}}
 \end{aligned}$$

The probable speed of the car before the breaks were applied, $u = 87.34 \text{ Kmph}$

$$1000 \text{ m} = 1 \text{ km}$$

$$m = \frac{1}{1000} \text{ km}$$

$$3600 \text{ Sec} = 1 \text{ hr}$$

$$\text{sec} = \frac{1}{3600} \text{ hr}$$

$$\frac{24.261 \times 1}{1000}$$

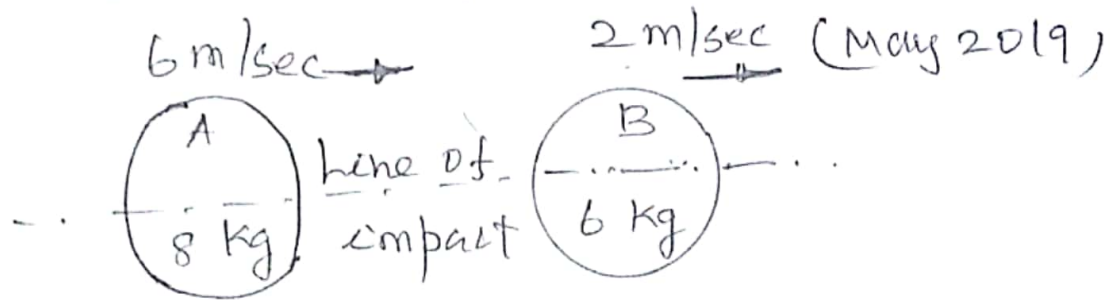
$$\frac{3600}{1} \times \frac{1}{3600}$$

$$= 24.261 \times 3600$$

$$= \frac{87.34 \text{ Kmph}}{1000}$$

Impact

1. Two bodies of 8 kg and 6 kg respectively move with velocities as shown in fig. Determine the velocity of each body directly after impact if the co-efficient of restitution is 0.6



Solution:

Given:

$$\left. \begin{array}{l} m_1 = 8 \text{ kg} \\ u_1 = 6 \text{ m/sec} \quad (\rightarrow) \end{array} \right\} \begin{array}{l} m_2 = 6 \text{ kg} \\ u_2 = 2 \text{ m/sec} \quad (\leftarrow) \\ e = 0.6 \end{array}$$

Principle of Conservation Momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$8 \times 6 + 6 \times 2 = 8 v_1 + 6 v_2$$

$$48 + 12 = 8 v_1 + 6 v_2$$

$$60 = 8 v_1 + 6 v_2$$

$\div$ by 2,

$$30 = 4 v_1 + 3 v_2$$

$$4 v_1 + 3 v_2 = 30 \quad \text{--- (1)}$$

co-efficient of restitution, $e = \frac{v_2 - v_1}{u_1 - u_2}$

$$0.6 = \frac{v_2 - v_1}{6 - 2}$$

$$0.6 = \frac{v_2 - v_1}{4}$$

$$2.4 = v_2 - v_1$$

$$-v_1 + v_2 = 2.4 \quad \text{--- (2)}$$

$$\textcircled{1} \Rightarrow 4v_1 + 3v_2 = 30$$

$$\textcircled{2} \times 4 \Rightarrow -4v_1 + 4v_2 = 9.6$$

$$7v_2 = 39.6$$

$$v_2 = \frac{39.6}{7} = \underline{\underline{5.657 \text{ m/sec}}}$$

sub. in $\textcircled{1}$,

$$4v_1 + 3(5.657) = 30$$

$$4v_1 + 16.97 = 30$$

$$4v_1 = 30 - 16.97 = 13.03$$

$$v_1 = \frac{13.03}{4} = \underline{\underline{3.257 \text{ m/sec}}}$$

2. A mass 12 kg travelling to the right with a speed of 7.5 m/sec collides with another mass 24 kg travelling to the left with a speed of 2.5 m/sec. If the coefficient of restitution is 0.6, Find the velocities of the particles after collision and loss of kinetic energy. What is the impulse acting on either particle during the impact?

Solution:

Given:

$$m_1 = 12 \text{ kg}$$

$$u_1 = 7.5 \text{ m/sec} \rightarrow$$

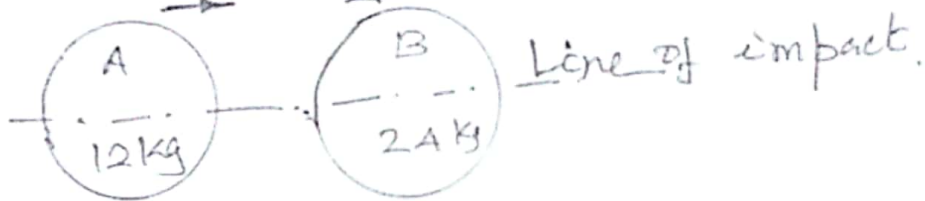
$$m_2 = 24 \text{ kg}$$

$$u_2 = -25 \text{ m/sec} \leftarrow$$

$$e = 0.6$$

$$u_1 = 7.5 \text{ m/sec} \rightarrow$$

$$u_2 = 25 \text{ m/sec} \leftarrow$$



$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

$$12(7.5) + 24(-25) = 12v_1 + 24v_2$$

$$90 - 600 = 12v_1 + 24v_2$$

$$-510 = 12v_1 + 24v_2$$

$$\div \text{ by } 12, \Rightarrow -42.5 = v_1 + 2v_2$$

$$v_1 + 2v_2 = -42.5 \quad \text{--- (1)}$$

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$$0.6 = \frac{v_2 - v_1}{7.5 + 25}$$

$$19.5 = v_2 - v_1$$

$$-v_1 + v_2 = 19.5 \quad \text{--- (2)}$$

$$\begin{aligned} \textcircled{1} + \textcircled{2} &\Rightarrow -v_1 + v_2 = 19.5 \\ v_1 + 2v_2 &= -42.5 \\ \hline 3v_2 &= -23 \end{aligned}$$

$$V_2 = -7.66 \text{ m/sec}$$

sub. in (1),

$$V_1 + 2(-7.66) = -42.5$$

$$V_1 = -27.16 \text{ m/sec}$$

$$\begin{aligned} \text{loss of kinetic Energy} &= \frac{1}{2}(m_1 V_1^2 + m_2 V_2^2) - \frac{1}{2}(m_1 u_1^2 + m_2 u_2^2) \\ &= \frac{1}{2}(12(-27.16)^2 + 24(-7.66)^2) - \frac{1}{2}(12(7.5)^2 + 24(25)^2) \\ &= 2705.56 \text{ J} \\ &= 2.705 \text{ KJ} \end{aligned}$$

Impulse:

$$\begin{aligned} \text{Impulse, } I &= \Sigma F \times t = m_1(V_1 - u_1) \\ &= 12(-27.16 - 7.5) \\ &= -415.92 \text{ J} \\ \therefore &= 415.92 \text{ J} \end{aligned}$$

Curvilinear Motion

1. A motorist is travelling on a curved portion of a highway radius 400m at speed of 90km per hour. The brakes are suddenly applied causing the speed to decrease at constant rate of 1.5 m/s^2 . Determine the magnitude of the acceleration of the automobile. Immediately after the brakes have been applied 3 sec. later. (Sep. 2020)

Solution:

Given

$$u = 90 \text{ kmph} = \frac{90 \times 10^3}{60 \times 60} = 25 \text{ m/s}$$

$$\text{radius, } r = 400 \text{ m,}$$

$$\text{retardation, } a_r = -1.5 \text{ m/s}^2$$

$$\text{time, } t = 3 \text{ s}$$

i) Normal acceleration of the automobile $a_n = \frac{v^2}{r}$

$$= \frac{(25)^2}{400} = \underline{\underline{1.56 \text{ m/s}^2}}$$

ii) Acceleration of the automobile after ~~brakes~~ brakes were applied, $a_r = -1.5 \text{ m/s}^2$ (given)

$$\begin{aligned} a &= \sqrt{(a_n)^2 + (a_r)^2} \\ &= \sqrt{(1.56)^2 + (-1.5)^2} \\ &= \underline{\underline{2.17 \text{ m/s}^2}} \end{aligned}$$

Acceleration of the automobile after 3 sec. later,

$$v = u + at$$

$$= 25 + (1.5)(3)$$

$$= \underline{20.5 \text{ m/sec}}$$

$$a_n = \frac{v^2}{r} = \frac{(20.5)^2}{400} = 1.05 \text{ m/sec}^2$$

$$a = \sqrt{(a_n)^2 + (a_r)^2}$$

$$= \sqrt{(1.05)^2 + (-1.5)^2} = \underline{1.83 \text{ m/sec}^2}$$

Relative velocity

1. Two ships move from a point at the same time. Ship A has a velocity of 30 kmph and is moving NW while ship B is moving in South-West direction with a velocity of 40 kmph. Determine the relative velocity of A with respect to B and distance between them after half an hour.

Solution;

$$V_{Ax} = 30 \cos 60^\circ = 15 \text{ kmph}$$

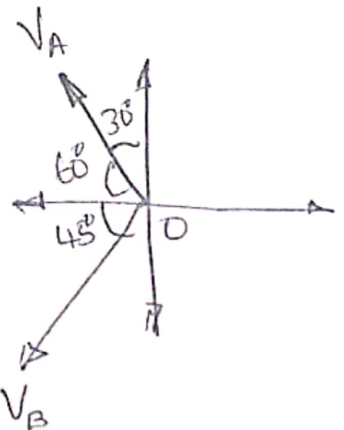
$$V_{Ay} = 30 \sin 60^\circ = 25.98 \text{ kmph}$$

$$V_{Bx} = 40 \cos 45^\circ = 28.284 \text{ kmph}$$

$$V_{By} = -40 \sin 45^\circ = -28.284 \text{ kmph}$$

$$V_{rx} = 15 - 28.284 = -13.284 \text{ kmph}$$

$$V_{ry} = 25.98 - (-28.284) = 54.264 \text{ kmph}$$



$$V_r = \sqrt{(13.284)^2 + (54.264)^2}$$

$$= 55.866 \text{ kmph.}$$

$$\tan \theta = \frac{13.284}{54.264}$$

$$\theta = 13.76^\circ$$

relative velocity of A with respect to B = 55.866 km

Relative distance after half an hour =

$$\text{Relative velocity} \times \text{time}$$

$$= 55.866 \times \frac{1}{2} = \underline{\underline{27.933 \text{ km}}}$$

Linear motion due to gravity

1. A stone is thrown vertically upwards with velocity of 30 m/s from the top of a tower 50 m high. Calculate:
 - (a) time required for the stone to reach the ground.
 - (b) velocity of the stone in its downward travel at the point of throwing upwards and
 - (c) maximum height to which the stone will rise in its flight.

Solution:

Given:

$$u = 30 \text{ m/s}$$

$$h = 50 \text{ m}, \quad v = 0$$

the motion of stone is the upward motion.

$$v^2 = u^2 + 2gh$$

$$0 = (30)^2 - 2 \times 9.81 h$$

$$2 \times 9.81 h = (30)^2$$

$$h = 45.87 \text{ m}$$

(c) The maximum height to which the stone rise $h = 45.87 \text{ m}$

(a) Time required for the stone to reach the ground (t),

$$t = t_1 + t_2$$

t_1 = time to reach upward motion

t_2 = time to reach downward motion

Upward motion:

$$v = u - gt_1$$

$$0 = 30 - 9.81 t_1$$

$$9.81 t_1 = 30$$

$$t_1 = \frac{30}{9.81} = \underline{\underline{3 \text{ sec}}}$$

Downward motion:

$$h = ut + \frac{1}{2}gt^2$$

$$45.87 = 0 + \frac{1}{2} \times 9.81 t_2^2$$

$$t_2^2 = 9.35$$

$$t_2 = \sqrt{9.35} = \underline{\underline{3 \text{ sec}}}$$

$$\therefore t = t_1 + t_2 = 3 + 3 = \underline{\underline{6 \text{ sec}}}$$

(b) The velocity of stone in the upward motion and downward motion are the same

$$v = \underline{\underline{30 \text{ m/sec}}}$$